

Impact Statement

Tokenizer for African Languages — Bridging the Tokenization Gap in African LLMs via SuperBPE — TRI AI Saturdays, Cohort 10

Purpose. This project builds and evaluates a SuperBPE tokenizer trained on genuine, multi-source text across nine African languages (eight originally targeted, plus Igbo as a documented bonus ninth language) spanning three major language families, addressing a documented “tokenization gap” in which standard, English-centric tokenizers fragment African-language words into small, often meaningless pieces. This inflates the computational cost of using AI systems in these languages and degrades model performance for their speakers — a concrete technical mechanism behind a broader pattern of digital exclusion.

Positive impacts. On held-out evaluation text drawn from all nine languages, the SuperBPE tokenizer reduced fragmentation by an average of 10.1% (tokens per word) relative to a matched standard byte-pair-encoding baseline, with every one of the nine languages showing improvement and no exceptions. An independent compression-efficiency metric (bytes per token) corroborated this with an average 11.8% improvement, converging on an identical per-language ranking to the fragmentation result — meaningful evidence that the effect is real rather than an artifact of either metric's definition. Gains were largest for Igbo, Hausa, and Yoruba (17–27% depending on the metric) and smallest for Amharic (2.3–2.4%), a pattern discussed further in the project's technical report. Lower token counts translate directly to lower compute cost and faster inference for any model built on this tokenizer, narrowing a real, measurable gap in what it costs to serve these languages compared to English.

Risks and harms. A more efficient tokenizer is a dual-use component: it could be adopted by well-resourced commercial actors as readily as by community-led or public-interest projects, so efficiency gains do not automatically translate into equitable access without attention to who deploys it and on what terms. There is also a risk that nine languages — a small and uneven slice of Africa's over 2,000 — could be mistaken for broad coverage, or inadvertently reinforce a hierarchy among African languages if presented without that caveat. At the small scale this project could train and evaluate (a roughly 9.4-million-parameter model, three training epochs), SuperBPE's tokenization advantage did not translate into better language-model quality — a genuine negative result reported openly rather than omitted. Training also required real, repeated compute time, a modest but real environmental cost worth naming.

Values. The project centers digital equity and linguistic inclusivity, and treats transparency as a working practice rather than a stated principle: every data source, every known limitation, and even a data-hygiene issue discovered after initial results were produced (see Limitations, below) are recorded rather than smoothed over.

Mitigation strategies. Uneven per-language data quality — retention after cleaning ranged from 27.0% (Zulu) to 93.4% (Igbo) across the nine languages — is disclosed in full in the project's Data Card rather than hidden, so downstream users can judge fitness for their specific language rather than assume uniform quality. Wolof's corpus was supplemented with a second, independent source (MasakhaNER 1.0) specifically because its Wikipedia-only corpus was flagged as insufficient before cleaning even confirmed it; sourcing decisions throughout favored properly licensed, encyclopedic, or research-community text (Wikipedia under CC-BY-SA-4.0, MasakhaNER under CC-BY-4.0-NC) over unlicensed convenience sources.

Limitations and uncertainties. This assessment reflects tokenizer-level metrics — compression, fragmentation, and losslessness — measured on held-out evaluation splits. Downstream effects on model fairness, representational bias, or real accessibility for these language communities remain untested, and would require further work with deployed language models and direct community input, not tokenizer statistics alone. A data-hygiene issue was also identified after these results were produced: the script that built the combined training corpus sourced from each language's full cleaned corpus rather than from the corpus with the evaluation split already excluded, meaning some held-out text may also have been present in training for all tokenizers evaluated. This is disclosed in full in the project's technical report; it does not change the relative comparison between the baseline and SuperBPE tokenizers, since both were affected identically, but it means the absolute figures reported above should be read with that caveat in mind pending a retrain on the corrected split.